

Introduction to Simulation

Contents

- Introduction
- Coins and Dice
- Queuing and Definitions

Introduction

Approaches

There are three “approaches” covered:

- By **hand** to gain understanding of the process
- Using **Excel** useful for small problems
- Using **Python** to code simulations, note you can use any other programming language

Stages in an Investigation

- How to construct a **simulation**
- Design of **experiments**
- **Analysis of results** from the **simulation**

Simulating Systems

Simulation techniques are used to enable the analysis of complex problems. Football fans, name a famous simulation?

Within Operational Research these problems often involve queuing applications

Questions to be answered:

- “can the current service system process the expected demand in a reasonable time?”
- “will the proposed new machinery improve (reduce) flow time or costs?”

Why Simulate?

- In most organisations it is not a feasible option that the firm carries out a full reorganisation (of its factory or sites of their factories)
- Simulation is cheap and can model complex interactions
- Allows prediction of future situations that have yet to occur
- Therefore, a firm can predict whether reorganisations are beneficial

Example

Considering a factory workshop the management might want to investigate

- Staffing levels, what is an ideal number and type of staff
- How many machines to use at each stage in the production process
- Should they use production lines
- What is an appropriate delivery/production time quotation (to a customer)

Evaluating Alternatives

When carrying out a Simulation there need to be some established criteria (to be able to evaluate alternatives)

Objective – e.g. would a new machine reduce costs and or improve performance

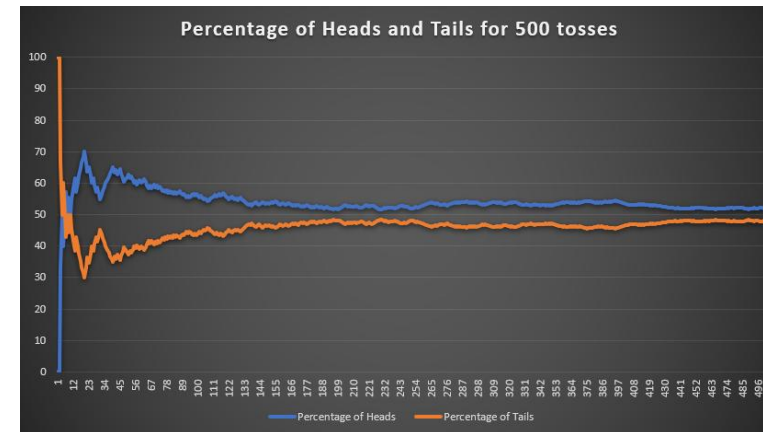
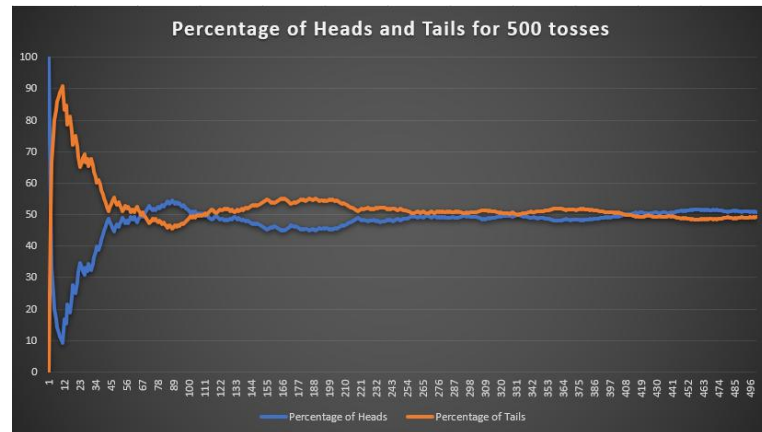
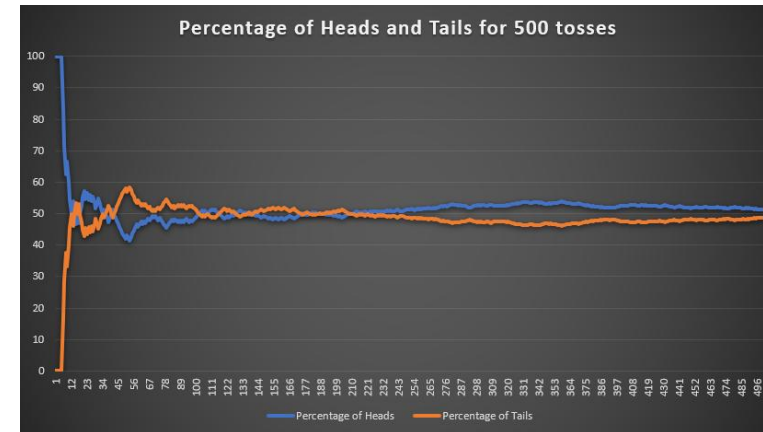
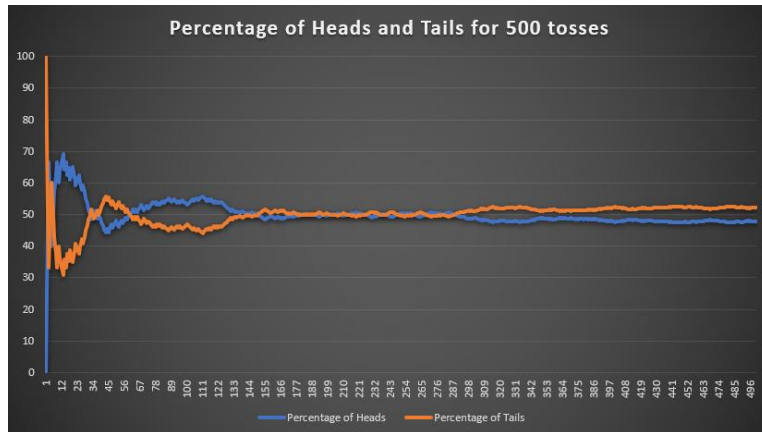
- Cost – e.g. Reduced cost through correct staffing or better machines – offset against investment needed for change
- Time – e.g. Reduced production time
- Time – e.g. Improved throughput, i.e. work in progress (including rework)

The Heated Clothes Rack

Coins and Dice

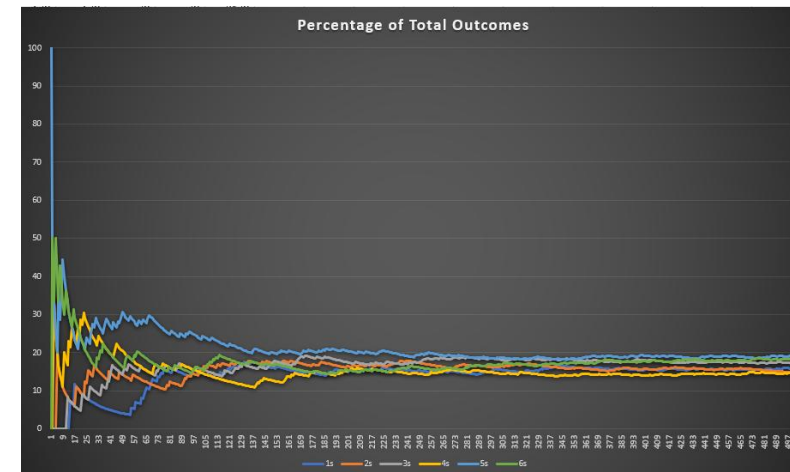
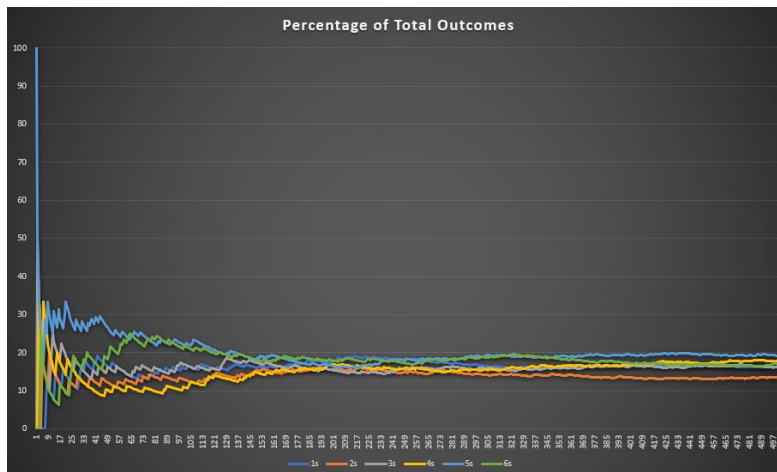
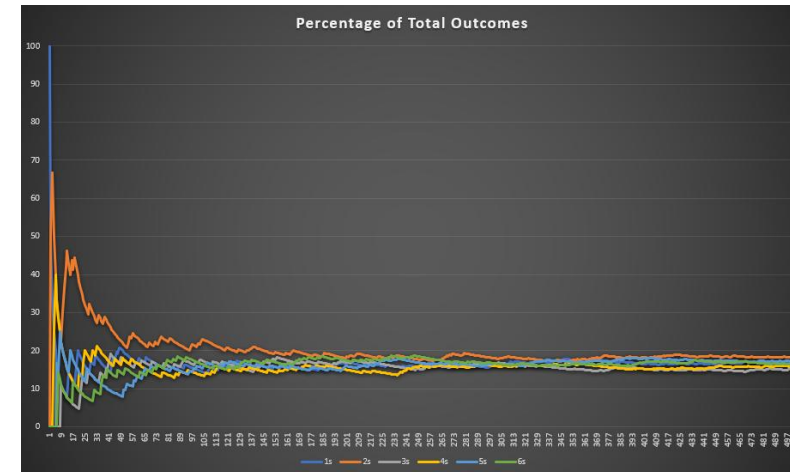
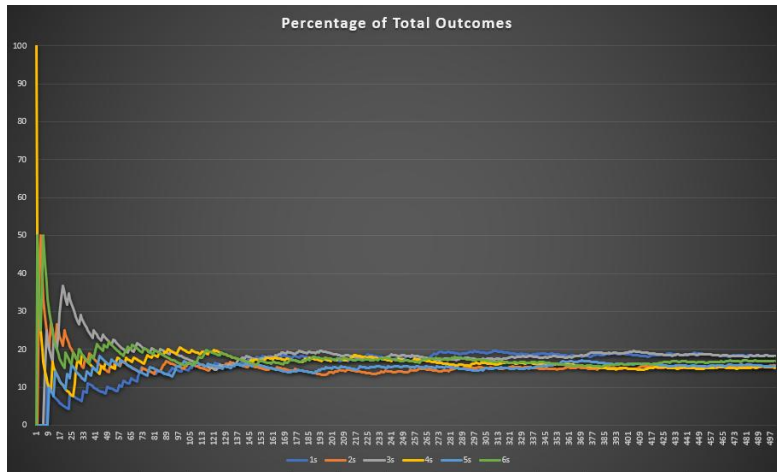
Simulating a Fair Coin Toss

Excel has been used to randomly simulate 500 trial tosses of a fair coin. 4 simulation runs are shown. In the long run we get roughly 50% heads. This is the law of large numbers at work.



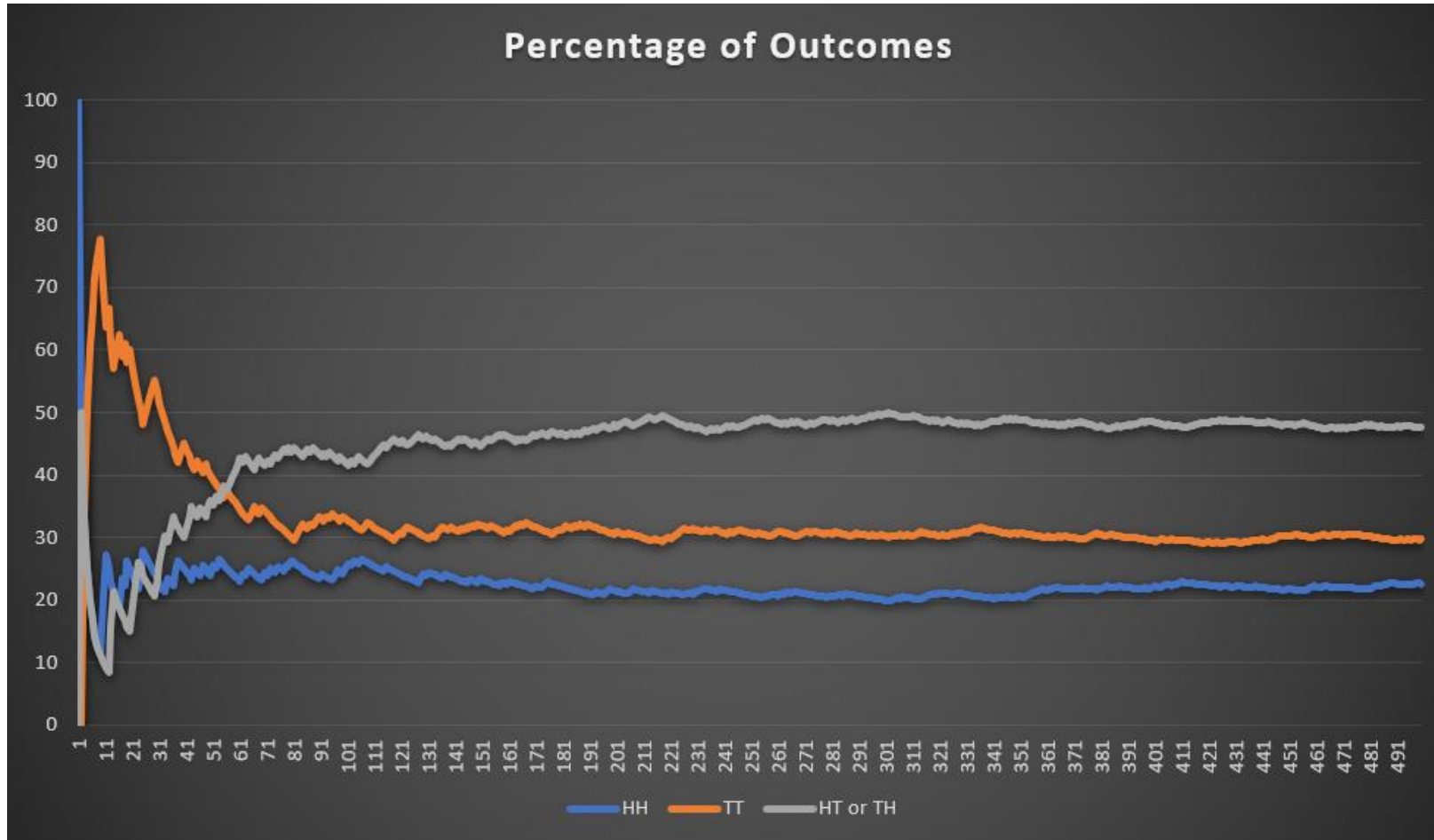
Simulating a Fair 6-Sided Dice

Excel has been used to randomly simulate the rolling of 6-sided dice for 500 trials. The 4 simulations are shown. In the long run we get roughly 16.67% of each number. This is the law of large numbers at work.



Simulating Two Fair Coin Tosses

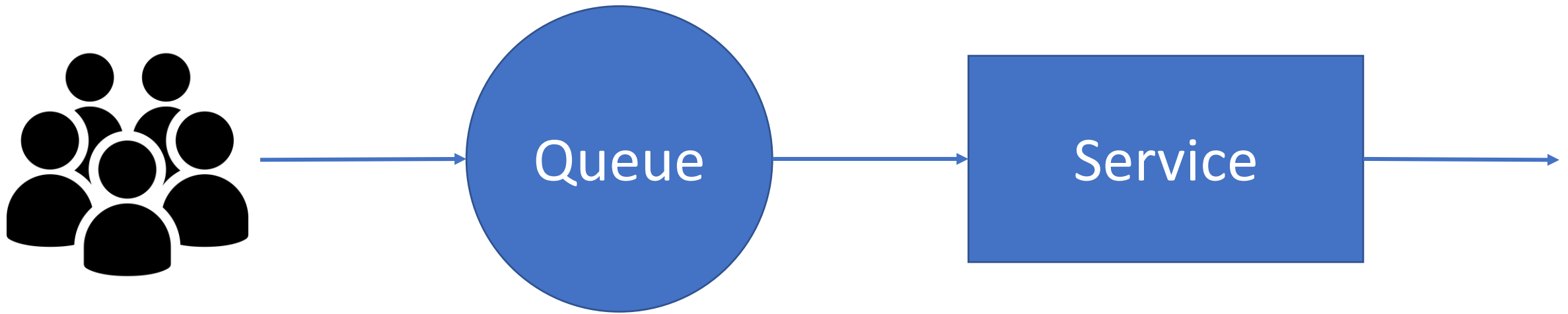
Excel has been used to randomly simulate 500 trials of 2 coin tosses of a fair coin.



Queuing and Definitions

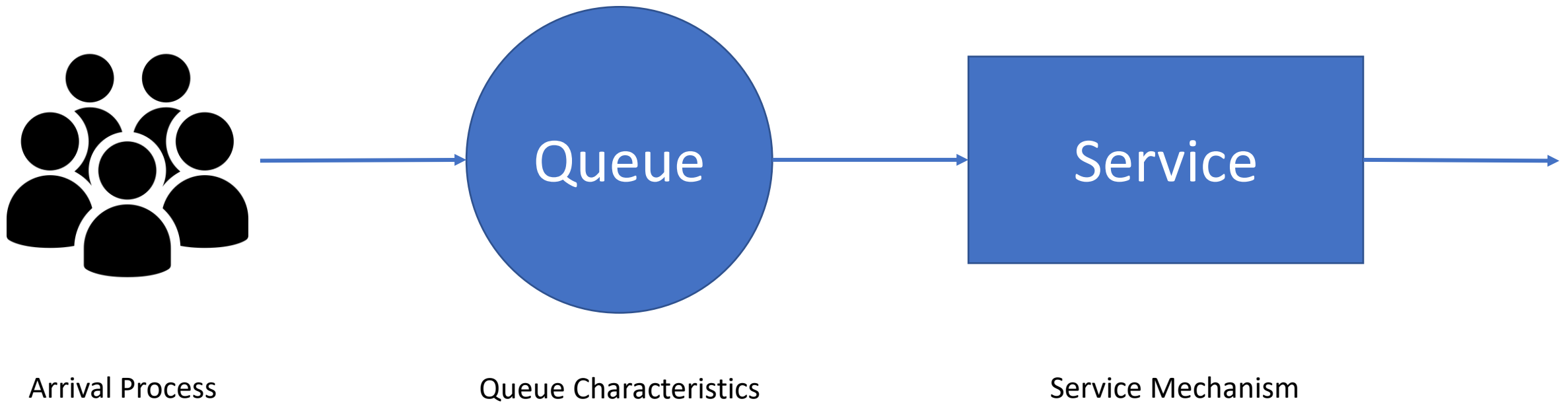
Queueing System

Customers queueing for a service, note this could be jobs waiting for a machine!

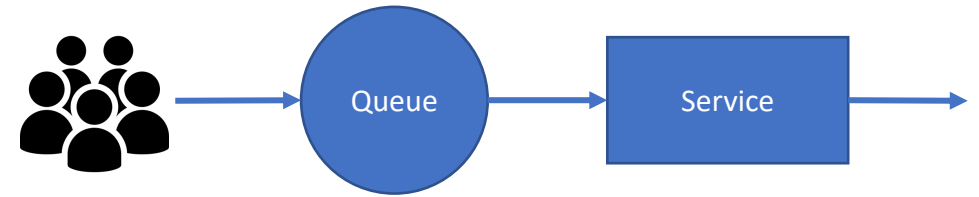


What are the unknowns?

Queueing System

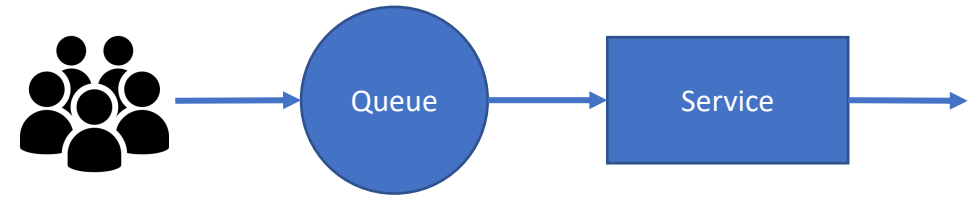


Arrival Process



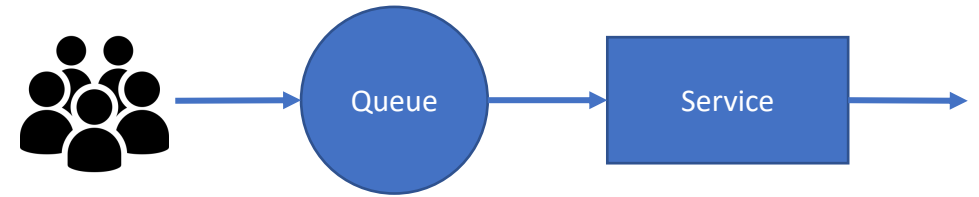
- how customers arrive e.g. singly or in groups (batch or bulk arrivals)
- how the arrivals are distributed in time (e.g. what is the probability distribution of time between successive arrivals (the ***inter-arrival time distribution***)
- whether there is a finite population of customers or (effectively) an infinite number

Queue Characteristics



- how, from the set of customers waiting for service, do we choose the one to be served next (e.g. FIFO (first-in first-out) also known as FCFS (first-come first served); LIFO (last-in first-out); randomly) (this is often called the *queue discipline*)
- do we have:
 - balking (customers deciding not to join the queue if it is too long)
 - reneging (customers leave the queue if they have waited too long for service)
 - jockeying (customers switch between queues if they think they will get served faster by so doing)
 - a queue of finite capacity or (effectively) of infinite capacity

Service Mechanism



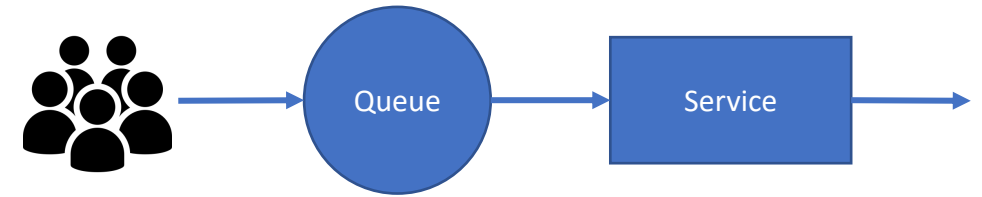
- a description of the resources needed for service to begin
- how long the service will take (the ***service time distribution***)
- the number of servers available
- whether the servers are in series (each server has a separate queue) or in parallel (one queue for all servers)
- whether pre-emption is allowed (a server can stop processing a customer to deal with another "emergency" customer)

Assumptions (initial)

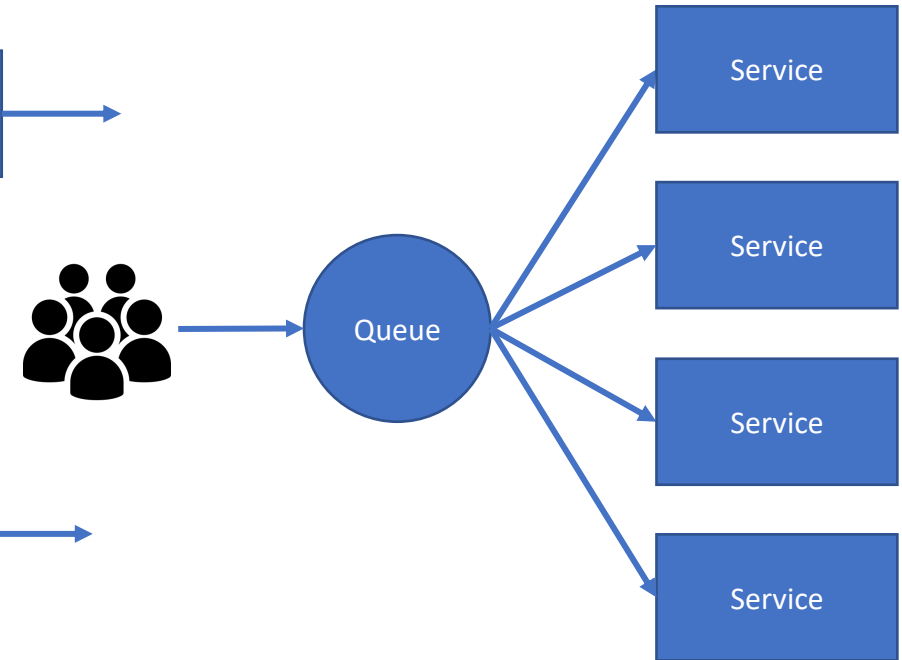
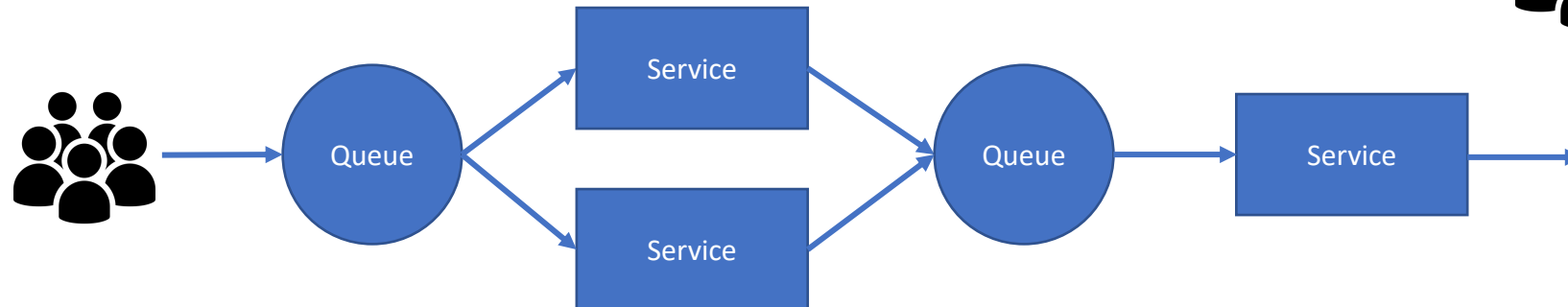
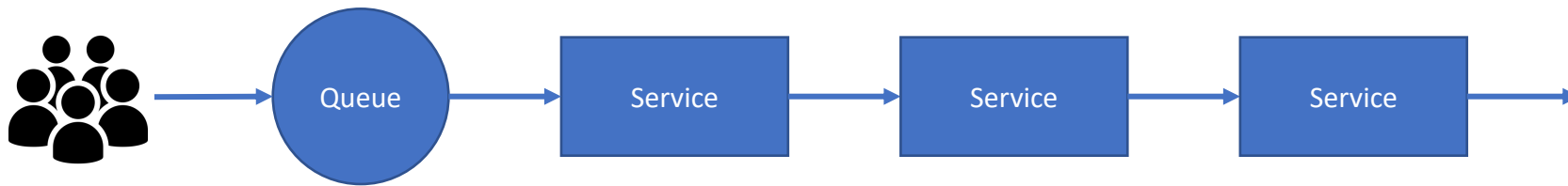
Standard assumptions are:

- All customers arrive singly
- Customers do not leave a queue once they have joined it (no reneging)
- Arriving customers do not refuse to enter a queue for any reason. (balking)
- One server can completely serve one customer

What can be simulated?

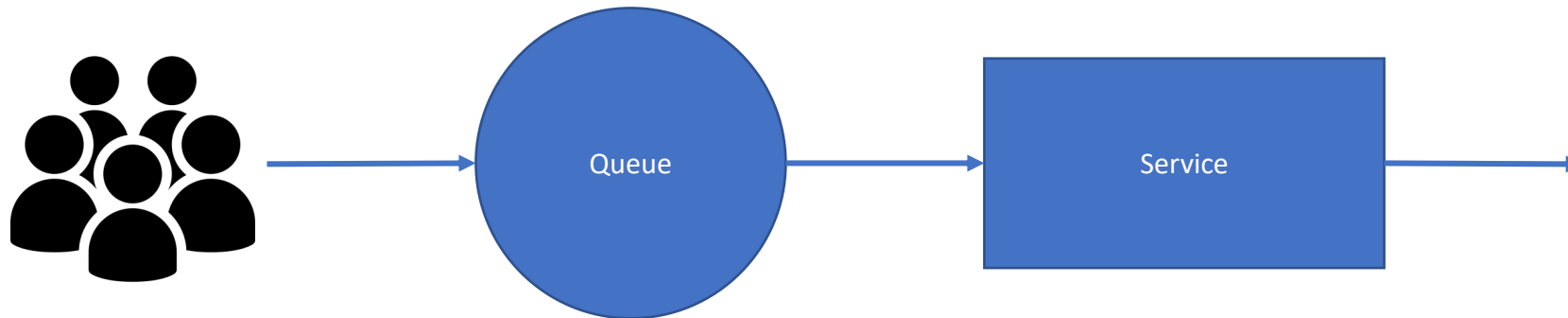


From the simple queueing process, we can chain these together in different configurations



Example – Doctor's Waiting Room

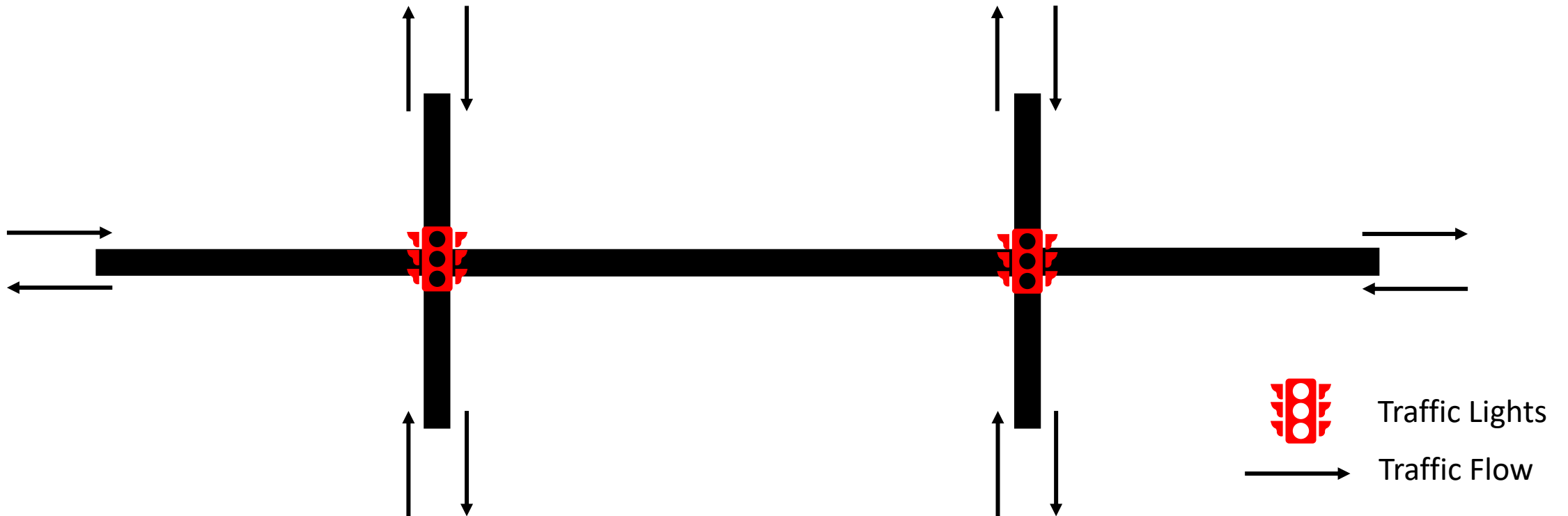
We've already seen the simple example. This is commonly referred to as the Doctor's waiting room



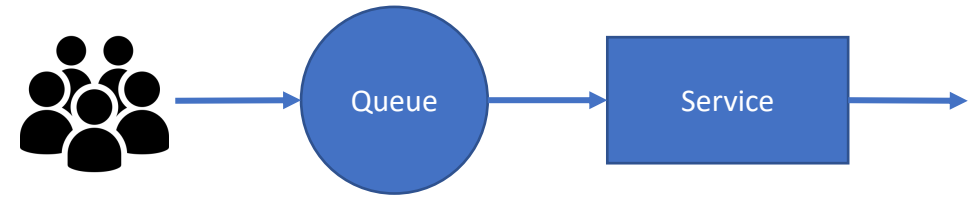
Example – Traffic Lights

What are the best light settings?

The aim of the simulation could be the determination of these settings.



Doctor's Waiting Room

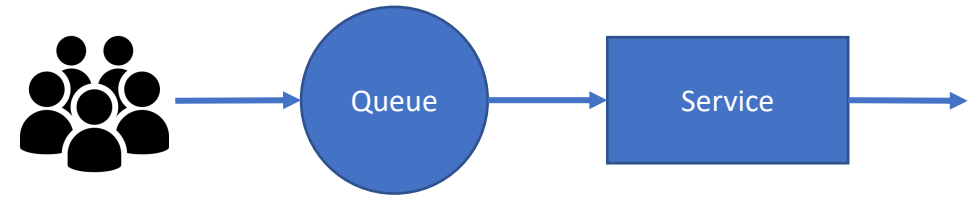


Here uncertainty in the system is caused by the variability in:

- Time between successive arrivals (***inter-arrival time***), and
- Time to serve a customer (***service time***).

Operating a Simulation

Operating a Simulation



To be able to represent an application, for example the doctor's waiting room (small shop), the following data needs to be defined:

- Arrival pattern/Arrival times
- Service rate/Service time

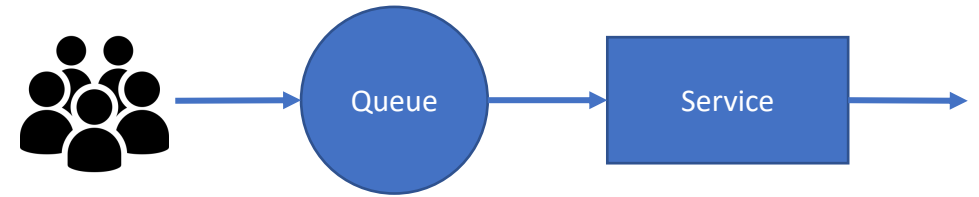
To illustrate the process assume that the first five jobs arrive at times

- 3, 8, 13, 19, 22

and that their respective service times are

- 4, 7, 3, 9, 11

Operating a Simulation



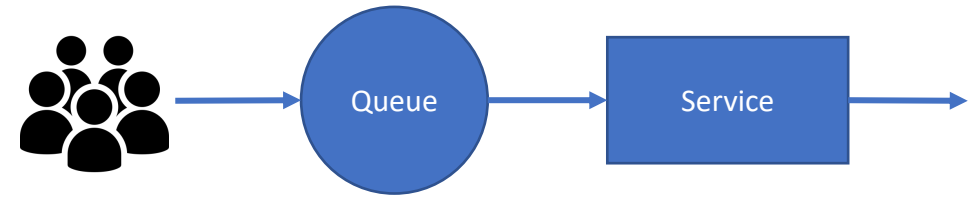
These values can be combined to carry out the simulation of this process to determine the simulation Metrics

- waiting time (customer)
- unused time (staff)

The “simulation” using these values is shown in the table on the next slide

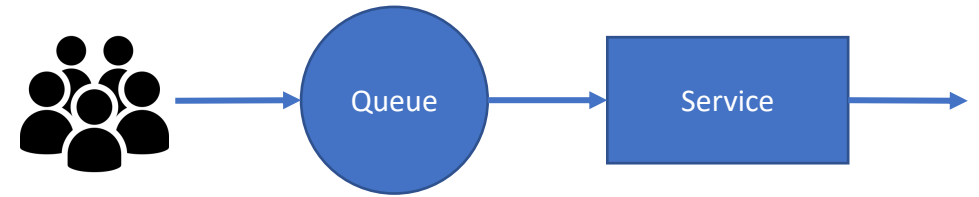
Values in black were given and those in red have been calculated

Operating a Simulation



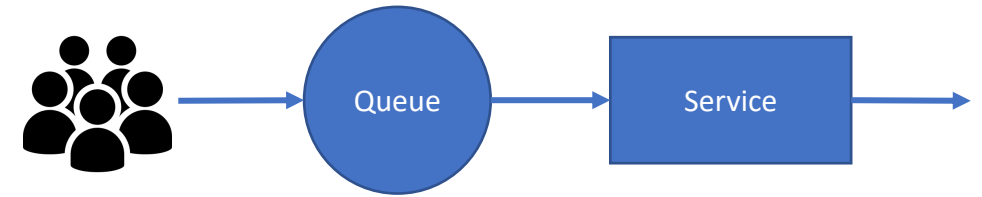
Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3		4			
8		7			
13		3			
19		9			
22		11			

Operating a Simulation



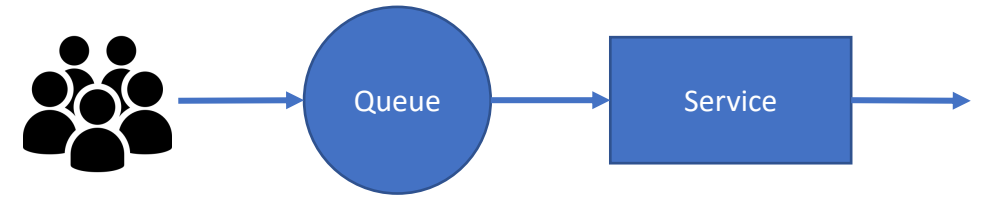
Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	$3+4=7$	$3-3=0$	3
8		7			
13		3			
19		9			
22		11			

Operating a Simulation



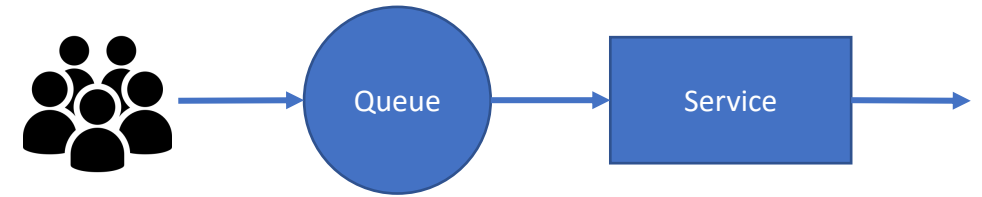
Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	7	0	3
8	$\max(8, 7) = 8$	7			
13		3			
19		9			
22		11			

Operating a Simulation



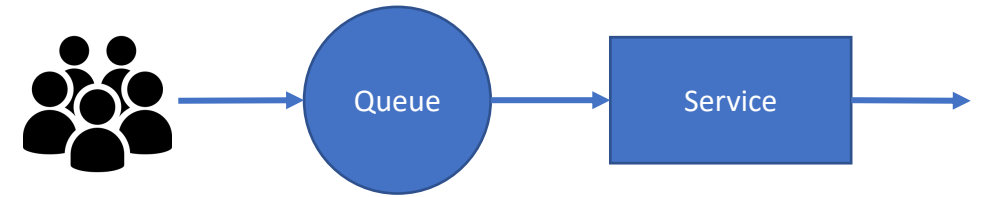
Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	7	0	3
8	8	7	8+7=15		
13		3			
19		9			
22		11			

Operating a Simulation



Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	7	0	3
8	8	7	15	8-8=0	
13		3			
19		9			
22		11			

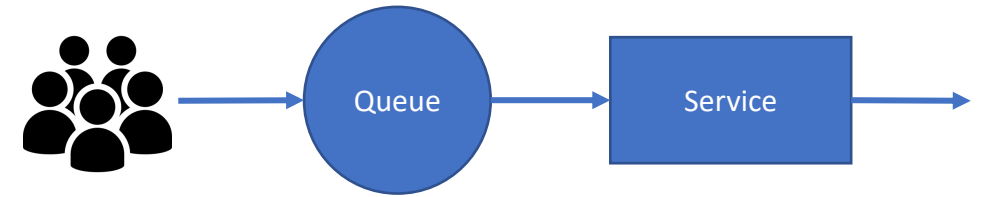
Operating a Simulation



Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	7	0	3
8	8	7	15	0	8-7=1
13		3			
19		9			
22		11			

Have a go at completing this table by hand. Solution on next slide

Operating a Simulation



Customer/Job Arrives	Service Starts	Service Duration	Service Ends	Customer/Job Waiting Time	Service Waiting Time
3	3	4	7	0	3
8	8	7	15	0	1
13	15	3	18	2	0
19	19	9	28	0	1
22	28	11	39	6	0

Simple Example

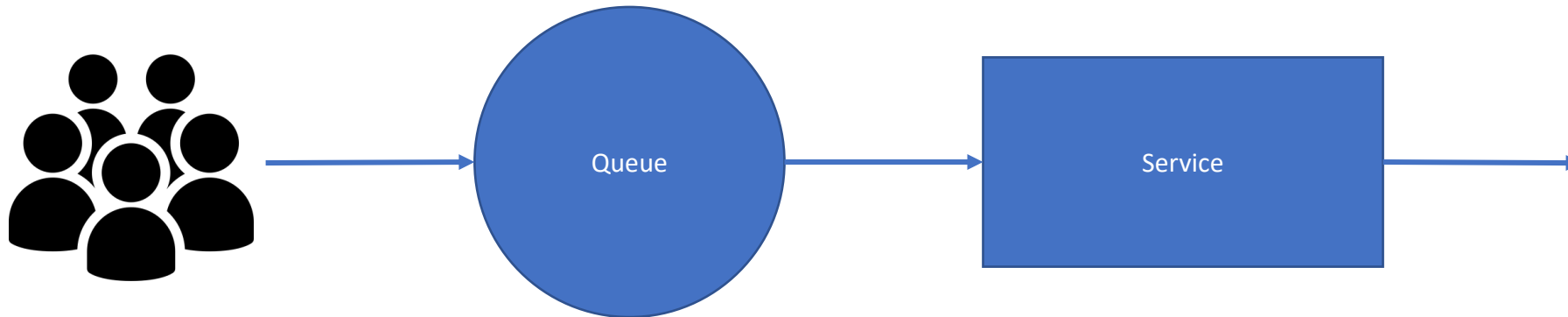
Operating a Simulation

Inter-arrival Time	8	12	16	20
Probability	0.2	0.4	0.2	0.2

Service Duration	7	11	15	18	20	22	24
Probability	0.3	0.2	0.1	0.15	0.15	0.05	0.05

This example uses

- 4 arrival times and their probability of occurrence
- 7 service duration times and their probability of occurrence



Inter-arrival Time	8	12	16	20
Cumulative Probability	0-0.2	0.2-0.4	0.6-0.8	0.8-1

Service Duration	7	11	15	18	20	22	24
Cumulative Probability	0-0.3	0.3-0.5	0.5-0.6	0.6-0.75	0.75-0.9	0.9-0.95	0.95-1

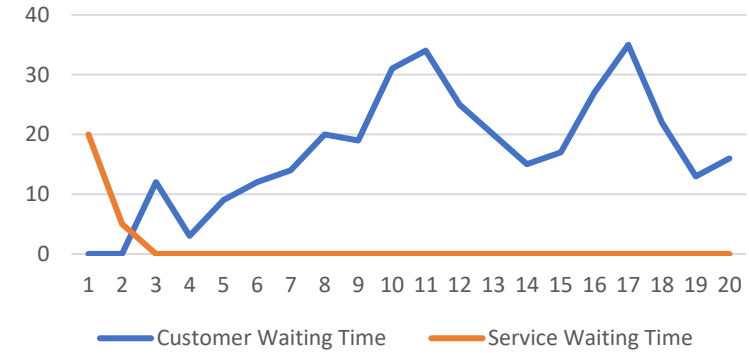
We can use the cumulative probability and randomly generate a number between 0 and 1 to pick a time for a customer and similarly generate their service time.

Operating a Simulation

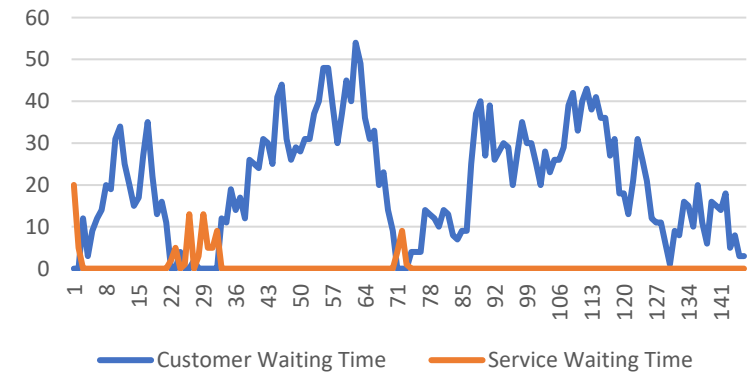
Inter-arrival Time	Service Duration
20	7
12	24
12	11
20	22
16	11
8	18
16	18
12	11
12	20
8	11
8	7
16	7
12	7
12	18
16	18

Customer Arrival Time	Service Starts	Service Duration	Service Ends	Customer Waiting Time	Service Waiting Time
20	20	7	27	0	20
32	32	24	56	0	5
44	56	11	67	12	0
64	67	22	89	3	0
80	89	11	100	9	0
88	100	18	118	12	0
104	118	18	136	14	0
116	136	11	147	20	0
128	147	20	167	19	0
136	167	11	178	31	0
144	178	7	185	34	0
160	185	7	192	25	0
172	192	7	199	20	0
184	199	18	217	15	0
200	217	18	235	17	0
208	235	20	255	27	0
220	255	7	262	35	0
240	262	7	269	22	0
256	269	15	284	13	0
268	284	7	291	16	0

Waiting Times



Waiting Times



Note with different inter-arrival times and service durations the waiting patterns would differ.

Basic Tips

Because a simulation is using randomness (stochasticity), it's behaviour need time to settle down. Think law of large numbers, flipping a coin.

- Run a simulation for a long enough time to be sure that the pattern you see isn't a fluke!
- Change some of your numbers to extremes and check that the simulation behaves reasonably.
 - E.g. Make the service time very large, we expect a queue to form for the customers